

Regression Assignmmt 2 Solutons

Student notes

1 Q1 Small training set idea

When training data is small then complex model is not good many time.

- Complex model $\rightarrow$ variance high and overfitting
- Simple model $\rightarrow$ little bias but ok prediction

Example : If only 10 house data points. High polynomial fits all but future wrong. Linear model simple but genralize better.

So simple model we choose.

2 Q2 Effect of lambda

Loss functon is

$$E = \sum [r_t - g(x_t|w)]^2 + \lambda \sum w_i^2$$

If λ big , weights small becomes.

Large $\lambda \rightarrow$ bias high
variance low

Small $\lambda \rightarrow$ bias low , variance high

3 Q3 Age and weight

Given values

$$n = 250$$

$$\sum x = 11211$$

$$\sum y = 44520.8$$

Mean calc

$$\bar{x} = \frac{11211}{250} = 44.844$$

$$\bar{y} = \frac{44520.8}{250} = 178.083$$

Now

$$S_{xx} = 543503 - 250(44.844)^2 = 40757$$

$$S_{xy} = 1996904.15 - 250(44.844)(178.083) = 414.15$$

Slope

$$b_1 = \frac{414.15}{40757} = 0.01016$$

Intercept

$$b_0 = 178.083 - 0.01016(44.844) = 177.63$$

Line is

$$\hat{y} = 177.63 + 0.01016x$$

At 25 age

$$\hat{y} = 177.63 + 0.254 = 177.88$$

Residual

$$e = 170 - 177.88 = -7.88$$

So over estimate it is.

4 Q4 Concrete data

Data solving gives

$$b_1 = -2.33$$

$$b_0 = 48.01$$

Model

$$\hat{y} = 48.01 - 2.33x$$

At $x = 4.3$

$$\hat{y} = 38$$

At $x = 3.7$

$$\hat{y} = 39.39$$

Residual

$$e = 46.1 - 39.39 = 6.71$$

Variance

$$\sigma^2 = 1.82$$

5 Q5 Multiple regression

Model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Normal eqns used and solve.

$$\beta_0 = 171.07$$

$$\beta_1 = 3.71$$

$$\beta_2 = -1.13$$

Predction

$$\hat{Y} = 171.07 + 3.71(18) - 1.13(43)$$

$$= 189.48$$

6 Q6 Body fat

Formula

$$\beta = (X^T X)^{-1} (X^T y)$$

Values get

$$\beta_0 = -6743.2$$

$$\beta_1 = -1715.8$$

$$\beta_2 = 1.79$$

Model

$$\%BF = -6743.2 - 1715.8H + 1.79W$$

7 Q7 Quadratic fit

Model is

$$f = w_0 + w_1x_1 + w_2x_2 + w_3x_1x_2 + w_4x_1^2 + w_5x_2^2$$

Matrix

$$\Phi = [1, x_1, x_2, x_1x_2, x_1^2, x_2^2]$$

Least square

$$w = (\Phi^T \Phi)^{-1} \Phi^T y$$

w_0 intercept

w_1, w_2 linear

w_3 mix term

w_4, w_5 curve part

These are quick notes only not full theory.